



LOGIX: AI-Powered Predictive Risk & Decision Intelligence for Global Supply Chains

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Problem Statement



PROBLEM

Global supply chains are highly interconnected, but disruption risk is difficult to quantify before it cascades.



Distributed disruptions: Port congestion, extreme weather, geopolitical events, strikes, customs bottlenecks and route blockages can affect multiple checkpoints.



Cascading impact: A disruption at one checkpoint can propagate through downstream transportation legs, increasing delay and delivery risk.



Limited decision-level visibility: Existing platforms provide strong tracking, predictive ETAs and disruption alerts, but there is an opportunity to better quantify checkpoint-level uncertainty, downstream propagation and deadline risk in one unified model.



High business cost: Uncertain delays can result in inventory shortages, production downtime, missed commitments, additional logistics costs and revenue loss.



Need for proactive decisions: Supply-chain teams need to know where risk originates, how far it may propagate, whether a delivery deadline is at risk, and which intervention or route can reduce that risk.



OBJECTIVE

Develop an **AI-powered early-warning and decision-support system** that models supply chains as **spatio-temporal checkpoint graphs**, predicts checkpoint-level **disruption risk and delay distributions**, propagates uncertainty to generate **probabilistic end-to-end ETAs**, identifies **critical bottlenecks**, and evaluates **alternative actions or routes** before disruptions become costly.



Solution

01 Data Collection & Integration

Collect shipment, route, port, weather, congestion, geopolitical, customs, and external event data.

02 Data Quality & Regime Detection

Clean data anomalies and identify genuine disruption and changing operational regimes.

03 Graph Construction & Route Mapping

Represent the shipment journey as connected checkpoints and transportation legs.

04 Feature Engineering & Context Fusion

Fuse temporal, operational, environmental, port, and external disruption signals.

05 Probabilistic Delay Forecasting

Estimate delay ranges using P50, P80, P90, and P95 predictions instead of a single value.



06 Spatio-Temporal Delay Propagation

Model how disruptions at one checkpoint cascade through downstream locations.

07 Monte Carlo Simulation

Simulate thousands of possible journeys to generate the end-to-end ETA distribution.

08 Risk & Attribution

Quantify deadline risk and identify the checkpoints and factors contributing most to the predicted delay.

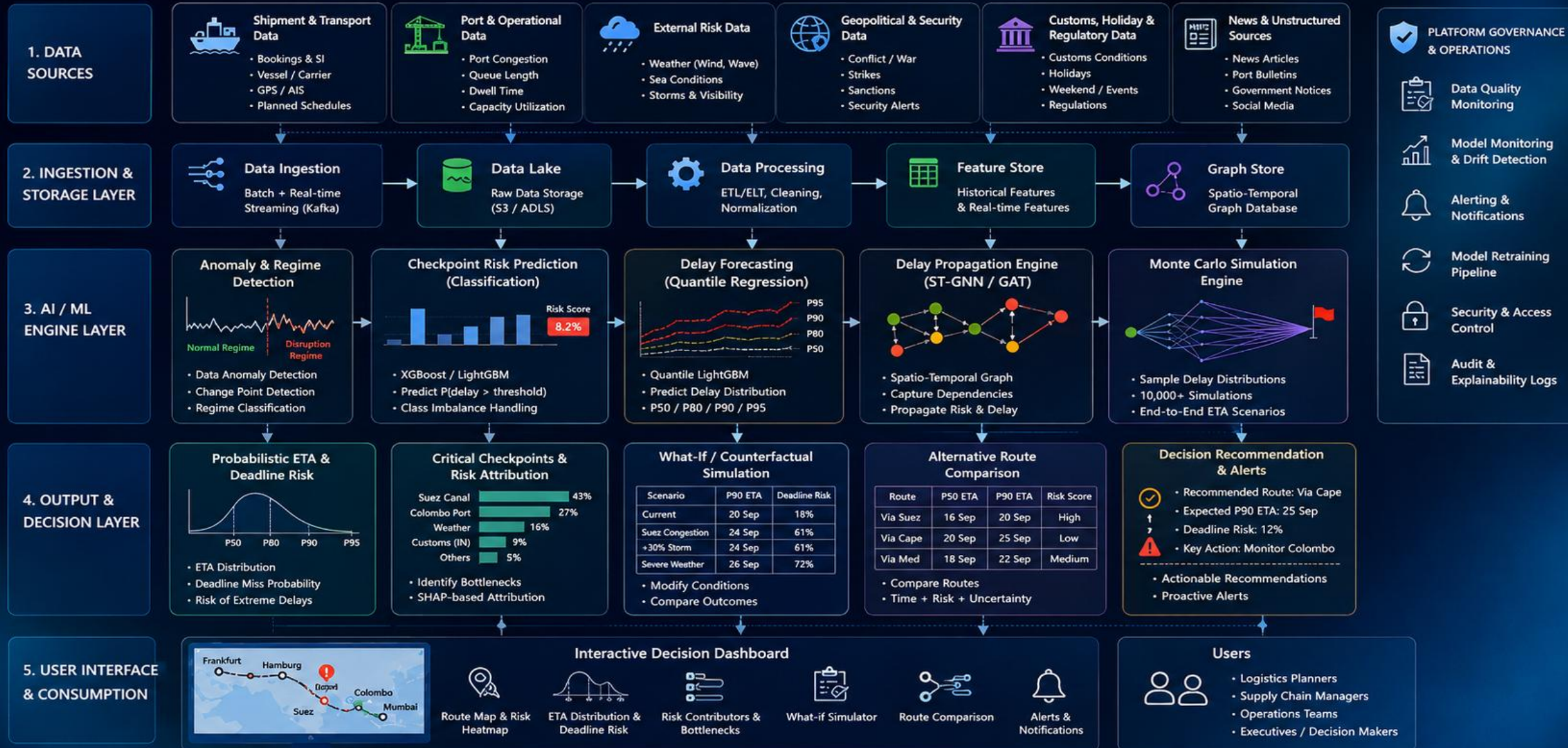
09 What-If Analysis

Simulate hypothetical disruptions and evaluate their impact on ETA and delivery risk.

10 Decision & Route Recommendation

Compare routes based on time, uncertainty, and disruption risk to recommend the best action.

Architecture



AI Component



AI Models

- Rule-based + Statistical Anomaly Detection
- Ruptures
- XGBoost / LightGBM
- LightGBM Quantile Regression
- GAT / ST-GNN
- SHAP + Graph Analysis
- Counterfactual + Monte Carlo
- NetworkX / OR-Tools
- Gemini / Groq LLM



Datasets

- NOAA MarineCadastre
- NOAA / NCEI
- ACLED
- GDELT / News Sources
- OpenStreetMap / Natural Earth
- Public Holiday Calendars / APIs
- Kaggle / Research Datasets



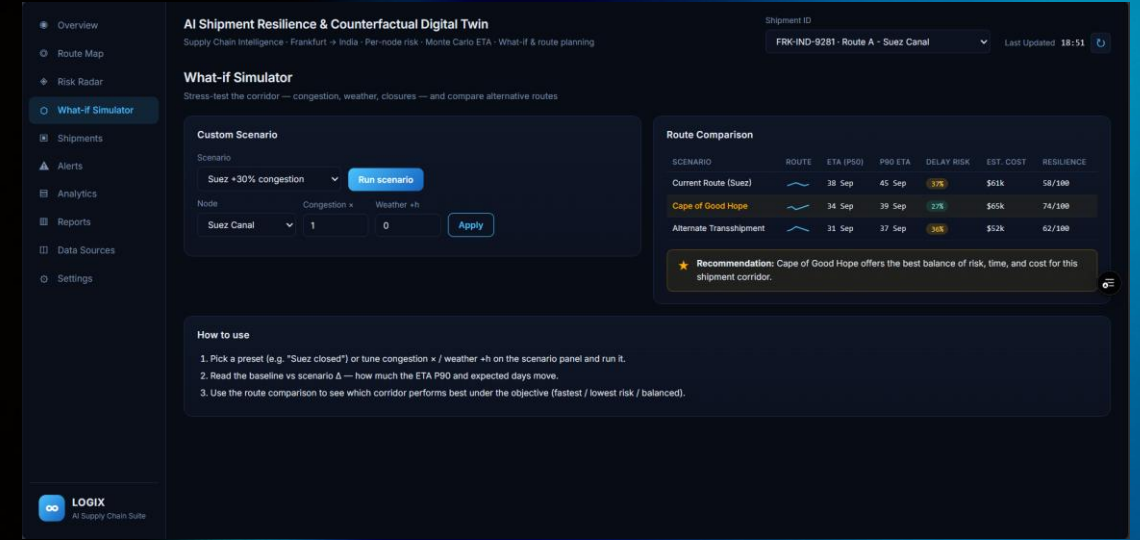
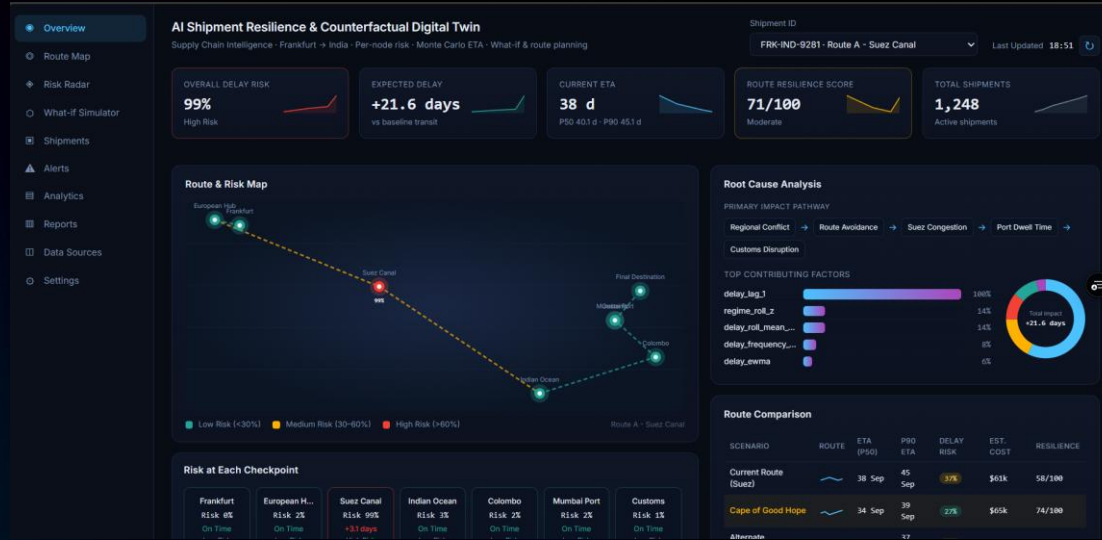
Tech Stack



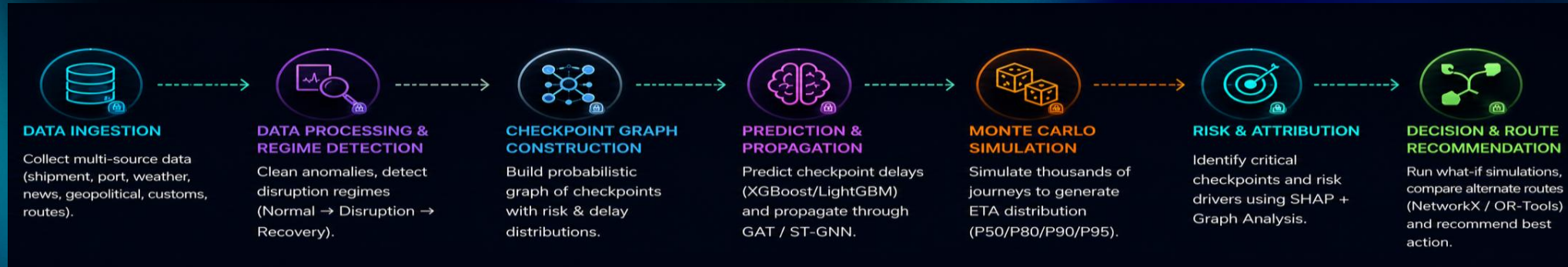
Prototype

[LIVE DEMO ↗](#)

Screenshots



Working Model – Process Flow



Innovation

1

Regime-Aware Disruption Intelligence



Detects **Normal** → **Disruption** → Recovery regimes and preserves major events such as canal blockages, wars, strikes, and congestion as predictive signals instead of treating them as ordinary outliers.

2

Probabilistic Checkpoint Graph



Represents the shipment as a network of **probabilistic checkpoints**, where every port, canal, customs point, and hub has its own evolving risk and delay distribution.

3

Spatio-Temporal Delay Propagation



Learns how disruptions **cascade** through connected checkpoints over time, estimating their downstream impact on the final delivery.

4

Critical Checkpoint & Uncertainty Attribution



Identifies which checkpoint and risk factor contribute most to final ETA uncertainty and deadline risk, answering “Where should we intervene?”

5

Counterfactual What-If Simulation



Allows decision-makers to **simulate interventions** such as increasing congestion or rerouting, then recomputes network risk, ETA distribution, and deadline probability.

6

Risk-Aware Route Decision



Selects routes using **transit time + uncertainty + disruption exposure + deadline risk**, rather than simply choosing the fastest or shortest route.

7

Dynamic Network Risk Recalculation



Automatically propagates new observations through the network, continuously updating **checkpoint risk → downstream risk → ETA → deadline probability**.

8

Explainable Network-Level Risk



Combines feature-level explanations with network-level impact, showing both **why** a checkpoint is risky and how much it affects the final ETA.

9

Causal / Intervention-Aware Reasoning



Extends prediction into intervention by modeling **potential cause-effect pathways** and evaluating alternative actions.

Impact



Expected Impact

- **Reduce ETA uncertainty** through probabilistic P50/P80/P90/P95 forecasts and deadline-miss probability.
- **Enable early disruption response** by identifying checkpoint-level risks before they cascade.
- **Reduce operational & financial losses** from delays, emergency rerouting, excess inventory, and missed commitments.
- **Improve routing decisions** by balancing transit time, disruption risk, uncertainty, and resilience.
- **Strengthen supply-chain resilience** by identifying vulnerable checkpoints and stress-testing alternative routes.



Target Users

Manufacturers • Logistics & Shipping Companies • Freight Forwarders •
Retailers • Supply-Chain Managers • Insurers



Business & Social Value

- Minimize cascading effects of transportation disruptions on businesses and consumers.
- Enable better utilization of transport capacity, fuel, inventory, and infrastructure.
- Improve reliability of customer commitments and critical deliveries.



Scalability

- **Multimodal:** Sea • Rail • Road • Air
- **Multi-Industry:** Automotive • Retail • Manufacturing • Pharma • Electronics
- **Global:** Scale from the Frankfurt → India corridor to interconnected global supply-chain networks.



Roadmap



Phase 1 — Data Foundation

Collect and integrate shipment, AIS, port, weather, geopolitical, customs, geographic, and disruption data.



Phase 2 — Data Processing & Feature Engineering

Clean anomalies, detect disruption regimes, and generate checkpoint- and route-level risk features.



Phase 3 — Predictive Intelligence

Build checkpoint delay classification and probabilistic delay forecasting using XGBoost/LightGBM and Quantile LightGBM.



Phase 4 — Graph Intelligence

Build the dynamic spatio-temporal supply-chain graph and implement GAT/ST-GNN for delay propagation.



Phase 8 — Product & Deployment

Build the interactive dashboard, API layer, visualization, alerts, and Dockerized deployment.



Phase 7 — AI & Real-Time Intelligence

Integrate news/event extraction and dynamic risk updates to continuously incorporate new disruption signals.



Phase 6 — Decision Intelligence

Add critical checkpoint attribution, what-if/counterfactual simulation, and risk-aware alternative-route comparison.



Phase 5 — Probabilistic ETA Engine

Run 10,000+ Monte Carlo simulations to generate P50/P80/P90/P95 ETA and deadline-miss probability.